

# Cristian F. López R.

(he/him/his)

September, 2026

## Postdoctoral Associate

Department of Applied Mathematics

University of Colorado Boulder

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[Personal website](#)

## EDUCATION

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| 2021 –<br>2025 | Ph.D. in Mechanical Engineering and Applied Mechanics<br>University of Nebraska – Lincoln (UNL), Lincoln, Nebraska, USA<br>Dissertation: “Data-driven Methods for Signal Analysis and Nonlinear System Identification”<br>M.S. in Mechanical Engineering and Applied Mechanics<br>Advisor: <a href="#">Keegan J. Moore</a> . |
| 2015 –<br>2017 | M.Eng. in Thermal Power Engineering<br>Zhejiang University (ZJU), Hangzhou, Zhejiang, China<br>Thesis: “Bearing Fault Diagnosis based on Improved Stochastic Resonance Method”<br>Advisor: <a href="#">Wei Zhong</a> .                                                                                                       |
| 2008 –<br>2013 | B.S. in Mechanical Engineering<br>Escuela Politécnica Nacional (EPN), Quito, Pichincha, Ecuador<br>Thesis: “The Design of a Heat Exchanger for the Cupola Furnace”<br>Advisor: <a href="#">Fausto Oviedo</a> .                                                                                                               |

## PROFESSIONAL EMPLOYMENT

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| 2025 –         | <b>Postdoctoral Associate</b> , Department of Applied Mathematics, University of Colorado Boulder, Boulder, Colorado, USA. Advisors: <a href="#">David Bortz</a> , <a href="#">Stephen Becker</a> . |
| 2022 –<br>2025 | <b>Research Assistant</b> , Department of Mechanical and Materials Engineering, UNL.                                                                                                                |
| 2021 –<br>2022 | <b>Teaching Assistant</b> , Department of Mechanical and Materials Engineering, UNL.                                                                                                                |
| 2018 –<br>2021 | <b>Lecturer</b> , Department of Basic Training, EPN.                                                                                                                                                |
| 2018           | <b>Lecturer</b> , Department of Industrial Engineering, Universidad Tecnológica Indoamérica, Quito, Ecuador.                                                                                        |
| 2015 –<br>2017 | <b>Research Assistant</b> , Department of Energy, ZJU.                                                                                                                                              |
| 2013 –<br>2014 | <b>Lecturer</b> , School of Mechanical Engineering, EPN.                                                                                                                                            |

## FUNDING

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| 2026 | Genesis: Transforming Science and Energy with AI. “ <a href="#">Weak-Form Agentic Digital Twins for Real-Time Control of Magnetically Confined Fusion Systems Under Sparse Observation</a> ”.<br>Awarded: \$750k.<br>Role: <b>Collaborator</b> . |
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## RESEARCH INTERESTS

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Weak form Scientific Machine Learning, Data-driven modeling and identification of dynamical systems, Model predictive control, Theoretical and experimental nonlinear dynamics and vibrations, Signal processing, Rotating machinery fault diagnosis.

## FELLOWSHIPS & AWARDS

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### Fellowships

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| 2021 – 2024 | 3. | Othmer Fellowship. UNL's graduate studies for academic merit.                        |
| 2021 – 2023 | 2. | Fulbright Faculty Development Program to study a Ph.D. program in the United States. |
| 2014 – 2018 | 1. | Chinese Government Scholarship. Language training and Master's studies.              |

### Awards

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| 2025 | 5. | Doctor Honoris Causa, Humanismus cum Rigoro Scientifico. Awarded by the Universidad GESTALT – Mexico; Ibarra – Ecuador. |
| 2023 | 4. | Travel Award, College of Engineering, UNL.                                                                              |
| 2022 | 3. | Travel Award, College of Engineering, UNL.                                                                              |
| 2017 | 2. | Best Presenter of the session, IEEE CSPA Conference, Penang, Malaysia.                                                  |
| 2013 | 1. | Academic Excellence, EPN. Top 2 % in the Mechanical Engineering program.                                                |

## PUBLICATIONS

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### Peer-Reviewed Journals

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| 2026 | 12. | <b>López C.</b> , Partidge M., De Pascuale S., Lore J., Christlieb A.J., Becker S., Bortz D.M., “ <a href="#">WSINDy for Model Predictive Control with Applications to Fusion, Drones, and Chaos</a> ”, <i>Proceedings of the Royal Society A</i> .                                    |
|      | 11. | Ghosh S., Andrulonis A., <b>López C.</b> , Singh A., Moore K.J., “ <a href="#">Energy-Based Dual-Phased Dynamics Identification for Forced Nonlinear Vibrations</a> ”, <i>Mechanical Systems and Signal Processing</i> .                                                               |
|      | 10. | <b>López C.</b> , Herzlieb J.E., Moore K.J., “ <a href="#">System Identification via Validation and Adaptation for Model Updating Applied to a Nonlinear Cantilever Beam</a> ”, <i>ASME Journal of Computational and Nonlinear Dynamics</i> .                                          |
|      | 9.  | <b>López C.</b> , Moore K.J., “ <a href="#">System Identification via Validation and Adaptation (SIVA) for Model Identification of Nonlinear Vibrating Structures</a> ”, <i>Nonlinear Dynamics</i> .                                                                                   |
| 2025 | 8.  | <b>López C.</b> , Naranjo Á., Salazar D., Moore K.J., “ <a href="#">Weak-form modified sparse identification of nonlinear dynamics</a> ”, <i>Journal of Computational Physics</i> .                                                                                                    |
|      | 7.  | <b>López C.</b> , Moore K.J., “ <a href="#">Energy-based dual-phase dynamics identification of clearance nonlinearities</a> ”, <i>Nonlinear Dynamics</i> .                                                                                                                             |
|      | 6.  | <b>López C.</b> , Singh A., Naranjo Á., Moore K.J., “ <a href="#">A data-driven, energy-based approach for identifying governing dynamics in vibrating structures directly from measurements</a> ”, <i>Mechanical Systems and Signal Processing</i> .                                  |
| 2024 | 5.  | <b>López C.</b> , Moore K.J., “ <a href="#">Enhanced adaptive linear chirplet transform for crossing frequency trajectories</a> ”, <i>Journal of Sound and Vibration</i> .                                                                                                             |
| 2022 | 4.  | <b>López C.</b> , Naranjo A., Lu S., Moore K.J., “ <a href="#">Hidden Markov model based stochastic resonance and its application to bearing fault diagnosis</a> ”, <i>Journal of Sound and Vibration</i> .                                                                            |
|      | 3.  | <b>López C.</b> , Wang D., Naranjo A., Moore K.J., “ <a href="#">Box-Cox-sparse-measures-based blind filtering: Understanding the difference between the maximum kurtosis deconvolution and the minimum entropy deconvolution</a> ”, <i>Mechanical Systems and Signal Processing</i> . |

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| 2017 | 2. <b>López C.</b> , Zhong W., Lu S., Cong F., Cortese I., “Stochastic resonance in an underdamped system with FitzHug-Nagumo potential for weak signal detection”, <i>Journal of Sound and Vibration</i> .<br>1. Zhong W., Chen X., Zhou Y., Wu Y., <b>López C.</b> , “Optimization of a solar aided coal-fired combined heat and power plant based on changeable integrate mode under different solar irradiance”, <i>Solar Energy</i> . |
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## Preprint

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| S | Ghosh S., <b>López C.</b> , Singh A., Moore K.J., “Decomposition-based Energy-based Dual-Phase Dynamics Identification for Nonlinear MDOF Systems”, <i>Mechanical Systems and Signal Processing</i> . |
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## CONFERENCE PRESENTATIONS & PROCEEDINGS

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| S    | 17. Ghosh S., <b>López C.</b> , Singh A., Moore K.J. “Energy-based Dual-phase Dynamics for Nonlinear System Identification of Multi-Degree-of-Freedom Systems”, <i>International Modal Analysis Conference (IMAC) XLV</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2026 | 16. Ghosh S., <b>López C.</b> , Singh A., Moore K.J. “Energy-based Dual-Phase Dynamics Identification For Multiple-Degree-of-Freedom Nonlinear Structures”, <i>NODYCON – The Fifth International Nonlinear Dynamics Conference</i> , Roma, Italy, September.<br>15. <b>López C.</b> , Becker S., Bortz D.M. “Data-Driven MPC via WSINDy for Plasma, Drone Control, and Chaotic Systems”, <i>Society for Industrial and Applied Mathematics – Annual Meeting</i> , Cleveland, OH, United States, July.<br>14. <b>López C.</b> , Moore K.J. “Data-driven Structural System Identification via Validation and Adaptation”, <i>International Modal Analysis Conference (IMAC) XLIV</i> , Palm Springs, CA, United States, January.<br>13. Ghosh S., Andrulonis A., <b>López C.</b> , Moore K.J. “A Data-driven Method for System Identification of Forced Nonlinear Vibrations Using Energy Conservation”, <i>IMAC XLIV</i> , Palm Springs, CA, United States, January.      |
| 2025 | 12. <b>López C.</b> , Moore K.J. “Generative Adversarial Networks with Embedded Physical Knowledge for Structural System Identification”, <i>International Design Engineering Technical Conferences &amp; Computers and Information in Engineering Conference (IDETC/CIE)</i> , Anaheim, CA, United States, August.<br>11. Ghosh S., Andrulonis A., <b>López C.</b> , Moore K.J. “Data-driven Identification of Forced Nonlinear Vibrations using Conservation of Energy”, <i>IDETC/CIE</i> , Anaheim, CA, United States, August.<br>10. <b>López C.</b> , Singh A., Naranjo A., Moore K.J. “A Data-Driven Methodology to Identify the Governing Dynamics in Vibrating Structures Through the Analysis of Mechanical Energy”, <i>IMAC XLIII</i> , Orlando, FL, United States, February.                                                                                                                                                                                  |
| 2024 | 9. <b>López C.</b> , Singh A., Naranjo A., Moore K.J. “A data-driven methodology to identify the governing dynamics in vibrating structures through the analysis of mechanical energy”, <i>European Nonlinear Oscillations Conference</i> , Delft, Netherlands, July.<br>8. Oliveira F., Johnson B., Singh A., Moore K.J. “Mathematical Modeling of the Nonlinear Dynamics of Bolted Joint Loosening”. <b>Presenter</b> . <i>European Nonlinear Oscillations Conference 2024</i> , Delft, Netherlands, July.<br>7. <b>López C.</b> , Ramsey T., Moore K.J. “Digital Engineering Test and Modeling Process: Autonomous Methods for Reconciling Test and Model Results”, <i>71st JANNAF Propulsion Meeting</i> , Oklahoma City, OK, United States, May.<br>6. <b>López C.</b> , Moore K.J. “Enhanced Adaptive Linear Chirplet Transform for Multi-Component Signals with Intersecting Instantaneous Frequencies”, <i>IMAC XLII</i> , Orlando, FL, United States, February. |

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|      | 5. | Arroyo J., Ramsey T., <b>López C.</b> , Moore K.J. “Learning Missing Data in Measurements using Meta-Modeling Techniques”, <i>IMAC XLII</i> , Orlando, FL, United States, February.                                                                                                                      |
| 2023 | 4. | <b>López C.</b> , Singh A., Naranjo A., Moore K.J. “Data-driven Euler-Lagrange approach for Time Series Analysis”, <i>IMAC XLI</i> , Austin, TX, United States, February.                                                                                                                                |
| 2022 | 3. | <b>López C.</b> , Naranjo A., Moore K.J. “Hidden Markov Model based Stochastic Resonance for bearing fault diagnosis”, <i>IMAC XL</i> , Orlando, FL, United States, February.                                                                                                                            |
| 2017 | 2. | <b>López C.</b> , Zhong W., Cong F., Hidalgo V., “ <a href="#">Bearing fault diagnosis based on Ensemble Empirical Mode Decomposition and Teager energy operator</a> ”, <i>IEEE 13th International CSPA</i> , Penang, Malaysia, March.                                                                   |
|      | 1. | <b>López C.</b> , Zhong W., Zheng M., “ <a href="#">Short-term Electric Load Forecasting Based on Wavelet Neural Network, Particle Swarm Optimization and Ensemble Empirical Mode Decomposition</a> ”, <i>Energy Procedia. 8th International Conference on Applied Energy</i> , Beijing, China, October. |

## WORKSHOPS

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| 2026 | 3. | Society of Industrial and Applied Mathematics Annual Meeting. Weak-Form Scientific Machine Learning: Theory and Applications Minisymposium. Cleveland, OH, USA. Jul. 6-10. Role: <b>Co-organizer</b> .                                       |
|      | 2. | WSINDy for model predictive control with applications to plasma boundary dynamics, drone tracking trajectory, and chaotic systems. Postdoctoral seminar, University of Colorado Boulder, Boulder, CO, USA. Apr. 29. Role: <b>Presenter</b> . |
| 2025 | 1. | Center for Hierarchical and Robust Modeling of Non-Equilibrium Transport Annual Meeting. University of Washington, Seattle, WA, USA. Dec. 1-3. Role: <b>Presenter</b> .                                                                      |

## INDUSTRY EXPERIENCE

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| 2022 –<br>2023 |  | Cruze Distribution LLC dba ARYSE under AFOSR STTR Phase II, “STTR: Performance Knee Brace”. Finite element models were used to simulate and experimentally test components of a new type of knee brace. |
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## TEACHING EXPERIENCE & ACTIVITIES

### University of Nebraska – Lincoln

#### Professional Development

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| 2025 | - UNL College of Engineering Graduate Student Teaching Fellow.                     |
|      | - Center for the Integration of Learning @ Nebraska Associate Level Certification. |

#### Course Taught

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|----------------|-------------------------------------------------------------|
| 2021 –<br>2022 | MENG 3250 Mechanics of Elastic Bodies (Teaching Assistant). |
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### Escuela Politécnica Nacional

#### Courses Taught

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|--------|--------------------------------------------------------------------------|
| 2018 – | - BAS010 Fundamentals of Physics                                         |
| 2021   | - FIS145 Newtonian Mechanics: Practical Component and FIS151 Laboratory. |

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|--------|----------------------------------------------------------------|
| 2013 – | - MEC1B2 Casting Technology                                    |
| 2014   | - MEC4E3 Numerical Methods                                     |
|        | - MEC4B4 Mechanics of Materials                                |
|        | - MEC7F4 Welding Production and Maintenance.                   |
| 2012 – | MEC1B1 Casting Laboratory (Laboratory Assistant and Auxiliar). |
| 2013   |                                                                |

## Universidad Tecnológica Indoamérica

### Courses Taught

|      |                              |
|------|------------------------------|
| 2018 | CAD/CAM/CIM and Maintenance. |
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## Mentoring

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| 2026 | 8. Tony Samour, University of Colorado Boulder, Master student.                                                                                    |
| 2025 | 7. Jackson E. Herzlieb, Georgia Institute of Technology, Undergraduate student.                                                                    |
| 2024 | 6. Alexis Andrulonis, Georgia Institute of Technology, Undergraduate student.                                                                      |
|      | 5. Sayantan Ghosh, UNL, Ph.D. student.                                                                                                             |
| 2023 | 4. Thomas Ramsey, UNL, Undergraduate student.                                                                                                      |
| 2022 | 3. Javier Arroyo, UNL, Ph.D. student.                                                                                                              |
| 2014 | 2. Sebastián Camacho. “Design and simulation of a clay modeling lathe”, EPN, Undergraduate student, committee member.                              |
|      | 1. Maritza Chuquirima and Lenin Cortez. “Study and Obtaining of Iron Metal from Ferruginous Sands”, EPN, Undergraduate students, committee member. |

## SERVICE

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| 2024   | 4. Certified Peer Reviewer: Elsevier, Researcher Academy.                                                                                                                         |
| 2023   | 3. Selection Committee – Fulbright Commission in Ecuador: Evaluation of applications.                                                                                             |
| 2022   | 2. Intermediary for Academic Collaboration: Facilitated research partnership between UNL’s Mechanical & Materials Engineering Department and EPN’s Mechanical Engineering School. |
| 2021 – | 1. Reviewer:                                                                                                                                                                      |
|        | <i>Journals</i>                                                                                                                                                                   |
|        | – Applied Mathematical Modelling                                                                                                                                                  |
|        | – Chaos, Solitons and Fractals                                                                                                                                                    |
|        | – Communications Physics                                                                                                                                                          |
|        | – Discover Applied Science                                                                                                                                                        |
|        | – IEEE Trans. on Industrial Electronics                                                                                                                                           |
|        | – IEEE Trans. on Signal Processing                                                                                                                                                |
|        | – Journal of Aircraft                                                                                                                                                             |
|        | – Journal of Nondestructive Evaluation                                                                                                                                            |
|        | – Journal of Signal, Image and Video Proc.                                                                                                                                        |
|        | – Journal of Sound and Vibration                                                                                                                                                  |
|        | – Mechanical Systems and Signal Proc.                                                                                                                                             |
|        | – Nonlinear Dynamics                                                                                                                                                              |
|        | – ROBOMECH Journal                                                                                                                                                                |
|        | – Scientific Reports                                                                                                                                                              |
|        | – Structural Health Monitoring                                                                                                                                                    |
|        | <i>Conference</i>                                                                                                                                                                 |
|        | International Design Engineering Technical Conferences & Computers and Information in Engineering Conference                                                                      |

## LANGUAGES

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Spanish (native), English (advanced), Mandarin (limited working proficiency).